

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 1 of 6

(1) Research Program/Project/Study Title: Product Development of Fruit and Vegetable Enriched Cassava Fries	
Proponent Name:	Joesan D. Borres, PhD-BM
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	Randolph V. Delfin (Asst. Prof. II)
Campus/Satellite College/Address:	Natividad, Pilar, Capiz
Telephone/Cell Phone/Fax Number:	
E-mail Address:	jdborres@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title:	1. Quality of Fruit and Vegetable Enriched Cassava Fries 2. Marketability of Fruit and Vegetable Enriched Cassava Fries 3. Feasibility of Fruit and Vegetable Enriched Cassava Fries 4. Shelf Life and Nutritional Adequacy of Fruit and Vegetable Enriched Cassava Fries 5. Social and Economic Impact of Production of Fruit and Vegetable Enriched Cassava Fries
Component Leader/s:	None
Academic Rank:	N/A

(2) Category (please check) ☐ Social Science/Education/Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research ☐ Developmental/Computer/Engineering/Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Pilar
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 2 of 6

(7) Sector/Commodity Food, Small Scale Entrepreneurs, Housewives, Kids,	(8) Discipline/Field of Specialization
(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.) Food trends come and go, but fries remain a culinary constant. The enduring popularity of the fries is fueled by its versatility and pairing ability as a perfect side dish. According to Technomic's recent <i>Starters, Small Plates & Sides Consumer Trend Report</i>, 64% of consumers said they would order fries as a lunch side and 58% would choose them as a side dish for dinner. And a majority of consumers (52%) said that if French fries were offered as an appetizer or small plate, they'd be likely to order them. Based from the prior art conducted, common fries existed in the market do not have much in the way of added ingredients. With this, the proponents are motivated to innovate fries existing in the market by producing cassava-potato fries, cassava-malunggay fries, cassava-carrot fries, cassava-squash fries, cassava-beerroot fries, cassava-dragon fruit fries, and cassava-mango fries. With that, offering this kind of products offer more nutritional value compared to ordinary potato fries.	
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)	
(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.) Generally, this study will be conducted to determine the quality of utilizing cassava into fries. Specifically, this study has the following objectives: 1. To determine the acceptability and sensory attributes in terms of appearance, color, taste, and texture of cassava-potato fries, cassava-malunggay fries, cassava-carrot fries, cassava-squash fries, cassava-beetroot fries, cassava-dragon fruit fries, and cassava-mango fries. 2. To determine the significant differences among treatment of fruit and vegetable enriched cassava fries. 3. To determine the comparative analysis of treatments among fruit and vegetable enriched cassava fries. 4. To determine the marketability among fruit and vegetable enriched cassava fries. 5. To determine the feasibility among fruit and vegetable enriched cassava fries. 6. To determine the shelf life and nutritional adequacy among fruit and vegetable enriched cassava fries. 7. To determine the social and economic impact of production of fruit and vegetable cassava fries.	
(11) Scientific Basis/Theoretical Framework (Include concepts. Theories and variables that indicate relevant relationships to be tested.	
(12) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology / information from which the project proposal will take-off). <u>CN105053898A</u> Making method of cassava whole flour chips. The invention discloses a making method of cassava whole flour chips. The making method of cassava whole flour chips comprises the following steps: pre-boiling cassava blocks, boiling, and processing to form mashed cassava; processing the mashed cassava through a noodle press and a knife to form cassava strips; and putting the cassava strips in an airfryer, coating with a layer of edible oil, frying at 180DEG C for 5min, overturning, and frying at 180DEG C for 5min. Pre-boiling keeps the integrity of cells as possible; boiling and mashing realize homogeneity of the appearance color and the mouthfeel of the cassava chips; pressing of the mashed cassava by the noodle press makes the cassava chips have a relatively homogenous thickness; the method reduces the use amount of the edible oil; and the method reduces the oil absorption amount of the cassava chips. The method is simple, greatly reduces the oil absorption rate of the cassava chips, and effectively reduces the over oxidation degree of oil in a novel heating mode; the novel heating mode makes the oil absorption rate of the cassava chips decrease to about 0.7% from 5.6%; and the edible oil is added several times, so multi-time over oxidation of oil at a high temperature is avoided.	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 3 of 6

CN104146226A Instant cassava whole-powder and production method thereof. The present invention discloses a production method for an instant cassava whole-powder. The method comprises the following steps: a, after peeling, cutting the cassava blocks to strips of 8-10 cm long, 0.4-0.6 cm thick by chip machine; b, placing the pre-boiled cassava strips into water at a temperature of 10 DEG C to cool for 30 minutes under continuous stirring; d, placing the cooled cassava strips in steam pot to carry out steam boiling for 35-40 minutes until the cassava strips are completely cooked; e, pounding the completely cooked cassava strips and carrying out sterilizing and drying in ohm cabinet drier for 8-12 minutes under continuous turning of the cassava strips; f, crushing the cassava strips after sterilizing and drying in step e into cassava powder on the superclean bench; g, passing the cassava powder in step f through 80 mesh steel sieve to get the cassava whole-powder, sealing and packaging for spare use. The cassava whole-powder produced by the present method has advantages of low iodine blue value, food safety, long shelf life and the like.

CN108432933A Manufacture method of preserved low-sugar fresh cassava. The present invention provides a manufacture method of preserved low-sugar fresh cassava. The manufacture method comprises the following steps: firstly, cassava tuberous roots are taken to be peeled; the peeled cassava tuberous roots are cut into strips to obtain cassava strips; the color-protecting, hardening and water bath treating are conducted; then sugar permeating treatment is conducted; and finally water washing and drying treatment are conducted to obtain the preserved low-sugar fresh cassava; the sugar permeating operation is as follows: sugar permeating liquid of 35-45% w/w of sucrose syrup and 0.15-0.25% w/w of citric acid is prepared: the water-bath treated cassava strips are placed in the sugar permeating liquid and put into a microwave oven at an output power of 250-300 w to be heated for 30-35 min; the heating is stopped for 10-15 min; and the steps of heating and non-heating are repeated for 3-4 times. The manufactured preserved low-sugar fresh cassava is uniform in the sugar permeating, low in sugar content, excellent in mouthfeel, and in line with modern healthy diet concepts, and belongs to the field of healthy food. The superior combination of ascorbic acid and the citric acid is used to effectively inhibit the browning of the cassava tuberous roots after harvesting. The microwave sugar permeating technology is used to enable the product to be even in the sugar permeating and greatly reduces the sugar content of the product. The preserved low-sugar fresh cassava is full in shape, short in time-consuming, low in sugar content and greatly improved in the overall quality.

CN101380088A Production method of cassava food. The invention relates to a method for producing a cassava food. The outer skin of cassava tuberous roots is peeled, and then the cassava tuberous roots are sliced, precooked, steamed with steam, dried and sieved to obtain finished products. The method has simple operation and low production cost; the finished products do not contain hydrocyanic acid, have the original flavor of cassava and can be eaten directly.

CN109497472A Preparation method of cassava chip. The invention discloses a preparation method of a cassava chip. The preparation method comprises the following steps of cleaning silt at the surface of cassava, boiling water, feeding the cassava, boiling for 15 to 20 min, peeling the skin, and soaking the cassava into normal-temperature clean water for 5 to 10 h; cutting the cassava into slices with thickness of 1.5 to 2 mm, and cleaning by clean water; adding 10 g of white sugar, 10 g of edible salt, 1 g of prickly ash, and 1 g of pepper powder into every 1000 g of water, boiling, feeding the cassava slices, and boiling for 2 to 4 min; fishing out the cassava slices, putting into the prepared ice water, cooling the cassava slices, and soaking into edible salt water with mass percentage of 1 to 4% for 10 to 15 min; fishing out the cassava slices, draining water, frying for 80 to 100s in edible oil, fishing out, airing to the room temperature, feeding the cassava slices into the edible oil, and frying again, so as to obtain the cassava chip. The prepared cassava chip is a novel cassava deep processing product, and has the advantages that by adding the white sugar, the edible salt, the prickly ash and other seasonings, the flavor is particular; after frying in oil twice, the cassava chip is crisp and tasty, and is more suitable for the taste of Chinese people.

ES2548277T3 Material de revestimiento para alimentos fritos. A coating material for fried foods and deep fried foods consisting of an oil/fat treated starch obtained by treating crude starch containing 50 mass % or more legume starch with ranges between 0, 01 and 1.0% by mass of an oil or fat, and matured by heat at a temperature of 30°C to 150°C for between 1 and 336 hours, the starch having an 3mylase content that varies between 10 and 60 % by mass as measured by amperometric titration according to the iodine affinity measurement method, characterized in that the oil/fat treated starch has a viscosity retention rate (V) of not less than 75% as calculated from the formula $V \geq (V_e/V_p) \times 100$ where V_p [BU] is the viscosity peak in a viscosity curve measured from a starch suspension having an anhydrous starch concentration of 8% of 40 °C to 95 °C at a rate of 1.5 °C/min, and if $n_d V_e$ [BU] the viscosity after a rest period of 10 min at 95 °C, with the condition that $V \geq 100 \%$ when no peak appears at the end of the measurement time.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 4 of 6

to be made, strategies for implementation.)

The main materials of cassava fries production such as a combination of cassava and potato, cassava and malunggay, cassava and carrot, cassava and squash, cassava and beetroot, cassava and dragon fruit, and cassava and mango. Other materials are filler and spices like eggs, flour, salt, pepper, sugar, milk, and cheese. Coconut oil is needed for frying.

The equipment to be used are knife, chopping board, measuring spoon, measuring cup, mixing bowl, casserole, bowl, tray, food processor, cling wrap, baking sheet, oven, deep frying pot, tongs, and paper towel.

The study will be used a single factor in a Complete Randomized Design (CRD) with five treatments replicated three times. A score sheet in 5-hedonic scale will be used to evaluate sensory characteristics of the product in terms of appearance, color, taste, texture, and acceptability. The product will be evaluated by 30 panelists who will be composed of selected 15 Faculty/Staff and 15 BSHM students.

(14) Major Activities/Work Plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.)

Research Objectives	Year 1		Year 2											
	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Write-Ups and Correction														
Acceptability and Sensory Evaluation														
Marketability Survey														
Financial Analysis														
Social and Economic Assessment														
Shelf Life Observation														
Nutritional Adequacy Testing														
Statistical Analysis														
Findings and Conclusions														
Presentation to Local and University RDE														
Application for IP protection/Journal Publication														

(15) Technology Transfer Activity Design(Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Fruit and Vegetable Enriched Cassava Fries IP Protection and Commercialization

(16) Targets Beneficiaries(Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Small Scale Entrepreneurs, Housewives, Kids

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Joesan D. Borres, PhD-BM	Study Leader	Business Management; Social Science
Randolph V. Delfin, MSBA	Co-Author	Hotel Restaurant and Management

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 5 of 6

Program/Project Members (in hours):

((18) Budgetary Requirement

Sources of Expenses	Amount (PhP)
A. Personnel Services	
Research Assistant	2,000.00
Statistician	2,000.00
English Critique	2,000.00
B. Maintenance and Other Operating	
Transportation Allowance	2,000.00
Meal allowance during data gathering	2,000.00
Ingredients	20,000.00
Shelf life and nutritional adequacy analysis	70,000.00
Equipment Needed	5,000.00
Incidental Expenses	5,000.00
TOTAL	110,000.00

(19) Literature Cited(include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

<https://ift.onlinelibrary.wiley.com/doi/abs/10.1111/j.1541-4337.2009.00077.x>
<https://www.foodservicedirector.com/menu/food-trends-come-go-fries-remain-culinary-constant>
<https://www.webmd.com/food-recipes/benefits-carrots#:~:text=The%20fiber%20in%20carrots%20can,are%20important%20for%20bone%20health.>
<https://www.sciencedirect.com/science/article/pii/S2213453016300362#:~:text=The%20leaves%20of%20M.,E%20also%20present%20in%20M.>
<https://foodrevolution.org/blog/health-benefits-of-squash/#:~:text=Squash%20are%20good%20sources%20of,and%20good%20for%20the%20gut.>
<https://foodrevolution.org/blog/health-benefits-of-squash/#:~:text=Squash%20are%20good%20sources%20of,and%20good%20for%20the%20gut.>
<https://www.webmd.com/food-recipes/benefits-dragon-fruit#:~:text=Dragon%20fruit%20is%20high%20in,in%20and%20use%20the%20iron.>
<https://worldwide.espacenet.com/patent/search?q=%22root%20crops%20chips%22>
<https://worldwide.espacenet.com/patent/search?q=nftxt%20%3D%20%22cassava%20chips%22%20AND%20nftxt%20%3D%20%22potato%20chips%22>
<https://worldwide.espacenet.com/patent/search?q=nftxt%20%3D%20%22cassava%20strips%22%20AND%20nftxt%20%3D%20%22potato%20strips%22>

(20) Capsule Curriculum Vitae(One-page brief C.V. only)

JOESAN D. BORRES

Instructor I/IPMO Campus Chair
 Capiz State University
 Pilar Satellite College

Educational Background

Doctor of Philosophy in Business Management (December 2020)
 Master of Arts in Teaching major in Social Science (October 2010)
 Master in Business Administration (March 2008)
 Bachelor of Science in Commerce major in Finance and Management Accounting (March 2005)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 6 of 6

Eligibilities

Licensed Professional Teacher
Civil Service Eligible (PD 907)
Bookkeeping NC III
Events Management NC III

RANDOLPH V. DELFIN

Asst. Prof. II/BSHM Program Chair
Capiz State University
Pilar Satellite College

Educational Background

Master of Science of Business Administration
Bachelor of Science in Hotel and Restaurant Management

Eligibilities

Licensed Professional Teacher
Events Management NC III

	Document Type:	FORM ISO 9001:2015	Document Code	IPMO-F04
			Revision No.	00
	Document Title:	PATENT SEARCH REPORT	Effective Date	September 16, 2019
			Page	1 of 3

University: CapSU – Pilar Satellite College
 ITSO Manager: Dr. Efren L. Linan
 ITSO Researcher: Dr. Joesan D. Borres

Annex - A SEARCH REPORT

Title: Product Development of Fruit and Vegetable Enriched Cassava Fries		
Filing Date November 9, 2022		
Abstract Food trends come and go, but fries remain a culinary constant. The enduring popularity of the fries is fueled by its versatility and pairing ability as a perfect side dish. Based from the prior art conducted, common fries existed in the market do not have much in the way of added ingredients. With this, the proponents are motivated to innovate fries existed in the market by producing cassava-potato fries, cassava-malunggay fries, cassava-carrot fries, cassava-squash fries, cassava-beetroot fries, cassava-dragon fruit fries, and cassava-mango fries. With that, offering this kind of products offer more nutritional value compared to ordinary potato fries.		
International Patent Classification A23		
Keywords <i>root crops fries, vegetable chips, cassava strips</i>		
Database	Search String	Number of Hits
espacenet	"cassava chips" AND "potato chips"	30
espacenet	"cassava strips" AND "potato strips"	3
espacenet	"root crops chips"	2

DOCUMENTS CONSIDERED TO BE RELEVANT	
Category*	Citation of Documents
A	CN105053898A Making method of cassava whole flour chips. The invention discloses a making method of cassava whole flour chips. The making method of cassava whole flour chips comprises the following steps: pre-boiling cassava blocks, boiling, and processing to form mashed cassava; processing the mashed cassava through a noodle press and a knife to form cassava strips; and putting the cassava strips in an airfryer, coating with a layer of edible oil, frying at 180DEG C for 5min, overturning, and frying at 180DEG C for 5min. Pre-boiling keeps the integrity of cells as possible; boiling and mashing realize homogeneity of the appearance color and the mouthfeel of the cassava chips; pressing of the mashed cassava by the noodle press makes the cassava chips have a relatively homogenous thickness; the method reduces the use amount of the edible oil; and the method reduces the oil absorption amount of the cassava chips. The method is simple, greatly reduces the oil absorption rate of the cassava chips, and effectively reduces the over oxidation degree of oil in a novel heating mode; the novel heating mode makes the oil absorption rate of the cassava chips decrease to about 0.7% from 5.6%; and the edible oil is added several times, so multi-time over oxidation of oil at a high temperature is avoided.
A	CN108432933A Manufacture method of preserved low-sugar fresh cassava. The present invention provides a manufacture method of preserved low-sugar fresh cassava. The manufacture method comprises the following steps: firstly, cassava tuberous roots are taken to be peeled; the peeled cassava tuberous roots are cut into strips to obtain cassava strips; the color-protecting, hardening and water bath treating are conducted; then sugar permeating treatment is conducted; and finally water washing and drying treatment are conducted to obtain the preserved

	Document Type:	FORM ISO 9001:2015	Document Code	IPMO-F04
			Revision No.	00
	Document Title:	PATENT SEARCH REPORT	Effective Date	September 16, 2019
			Page	1 of 3

	low-sugar fresh cassava; the sugar permeating operation is as follows: sugar permeating liquid of 35-45% w/w of sucrose syrup and 0.15-0.25% w/w of citric acid is prepared: the water-bath treated cassava strips are placed in the sugar permeating liquid and put into a microwave oven at an output power of 250-300 w to be heated for 30-35 min; the heating is stopped for 10-15 min; and the steps of heating and non-heating are repeated for 3-4 times. The manufactured preserved low-sugar fresh cassava is uniform in the sugar permeating, low in sugar content, excellent in mouthfeel, and in line with modern healthy diet concepts, and belongs to the field of healthy food. The superior combination of ascorbic acid and the citric acid is used to effectively inhibit the browning of the cassava tuberous roots after harvesting. The microwave sugar permeating technology is used to enable the product to be even in the sugar permeating and greatly reduces the sugar content of the product. The preserved low-sugar fresh cassava is full in shape, short in time-consuming, low in sugar content and greatly improved in the overall quality.
A	CN104146226A Instant cassava whole-powder and production method thereof. The present invention discloses a production method for an instant cassava whole-powder. The method comprises the following steps: a, after peeling, cutting the cassava blocks to strips of 8-10 cm long, 0.4-0.6 cm thick by chip machine; b, placing the pre-boiled cassava strips into water at a temperature of 10 DEG C to cool for 30 minutes under continuous stirring; d, placing the cooled cassava strips in steam pot to carry out steam boiling for 35-40 minutes until the cassava strips are completely cooked; e, pounding the completely cooked cassava strips and carrying out sterilizing and drying in ohm cabinet drier for 8-12 minutes under continuous turning of the cassava strips; f, crushing the cassava strips after sterilizing and drying in step e into cassava powder on the superclean bench; g, passing the cassava powder in step f through 80 mesh steel sieve to get the cassava whole-powder, sealing and packaging for spare use. The cassava whole-powder produced by the present method has advantages of low iodine blue value, food safety, long shelf life and the like.
A	CN101380088A Production method of cassava food. The invention relates to a method for producing a cassava food. The outer skin of cassava tuberous roots is peeled, and then the cassava tuberous roots are sliced, precooked, steamed with steam, dried and sieved to obtain finished products. The method has simple operation and low production cost; the finished products do not contain hydrocyanic acid, have the original flavor of cassava and can be eaten directly.
A	CN109497472A Preparation method of cassava chip. The invention discloses a preparation method of a cassava chip. The preparation method comprises the following steps of cleaning silt at the surface of cassava, boiling water, feeding the cassava, boiling for 15 to 20 min, peeling the skin, and soaking the cassava into normal-temperature clean water for 5 to 10 h; cutting the cassava into slices with thickness of 1.5 to 2 mm, and cleaning by cleanwater; adding 10 g of white sugar, 10 g of edible salt, 1 g of prickly ash, and 1 g of pepper powder into every 1000 g of water, boiling, feeding the cassava slices, and boiling for 2 to 4 min; fishing out the cassava slices, putting into the prepared ice water, cooling the cassava slices, and soaking into edible salt water with mass percentage of 1 to 4% for 10 to 15 min; fishing out the cassava slices, draining water, frying for 80 to 100s in edible oil, fishing out, airing to the room temperature, feeding the cassava slices into the edible oil, and frying again, so as to obtain the cassava chip. The prepared cassava chip is a novel cassava deep processing product, and has the advantages that by adding the white sugar, the edible salt, the prickly ash and other seasonings, the flavor is particular; after frying in oil twice, the cassava chip is crisp and tasty, and is more suitable for the taste of Chinese people.
A	ES2548277T3 Material de revestimiento para alimentos fritos. A coating material for fried foods and deep fried foods consisting of an oil/fat treated starch obtained by treating crude starch containing 50 mass % or more legume starch with ranges between 0, 01 and 1.0% by mass of an oil or fat, and matured by heat at a temperature of 30°C to 150°C for between 1 and 336 hours, the starch having an amylase content that varies between 10 and 60 % by mass as measured by amperometric titration according to the iodine affinity measurement method, characterized in that the oil/fat treated starch has a viscosity retention rate (V) of not less than 75% as calculated from the formula $V \geq (V_e/V_p) \times 100$ where V_p [BU] is the viscosity peak in a viscosity curve measured from a starch suspension having an anhydrous starch concentration of 8% of 40 °C to 95 °C at a rate of 1.5 °C/min, and if V_e [BU] the viscosity after a rest period of 10 min at 95 °C, with the condition that $V \geq 100$ % when no peak appears at the end of the measurement time.

	Document Type:	FORM ISO 9001:2015	Document Code	IPMO-F04
			Revision No.	00
	Document Title:	PATENT SEARCH REPORT	Effective Date	September 16, 2019
			Page	1 of 3

*** Special Categories of Cited Documents:**

- A** document defining the general state of the art which is not considered to be of particular relevance
- X** document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
- Y** document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

Note: Further references not indicated in this search report may be cited during substantive examination.

3. CERTIFICATION

Signature of Authorized Officer